Applied Deep Learning

Multi-Class Segmentation of Tree Species Using UAV Imagery and U-Net

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Abstract: - This project applies a U-Net–based convolutional neural network for multi-class semantic segmentation of tree species in high-resolution UAV orthomosaic imagery. A custom RGB dataset was created using a publicly available orthomosaic. Tree crowns were manually delineated in QGIS, followed by tiling and mask generation. The initial six-class model (including background and five species) achieved a macro mean Intersection over Union (IoU) of ~0.36, even with Albumentations-based augmentation. Simplifying the task to five classes (background, beech, pine, birch, spruce) improved performance, with the final U-Net trained for 30 epochs achieving a macro mean IoU of ~0.48 and a macro mean Dice coefficient of ~0.56. Results demonstrate that deep learning can identify major tree species from UAV imagery with modest models and limited data, but challenges remain for rare or visually similar species.

Keywords: - Applied Deep learning, semantic segmentation, U-Net, UAV imagery, tree species mapping

1. Introduction

Tree species information is crucial for forest management, biodiversity monitoring and climate-related risk assessment, yet traditional field-based inventories are costly and spatially restricted despite their high accuracy at tree level (Qin et al. 2022). High-resolution UAV imagery offers an attractive alternative by providing detailed crown-scale information at comparatively low cost, but automatic tree species identification remains challenging due to complex canopy structures, variable illumination conditions and high intra-class variability in crown appearance (Qin and Zhao 2025).

Semantic segmentation with convolutional neural networks (CNNs) has become a standard paradigm for pixel-wise classification in remote sensing, enabling dense labelling of vegetation and land-cover classes (Zhao et al. 2025). Encoder–decoder architectures such as U-Net are particularly suitable for tasks requiring precise boundary delineation, as skip connections fuse fine-scale texture information from shallow layers with broader spatial context from deeper layers, and have been successfully applied beyond their original biomedical domain to vegetation and tree species mapping (Ronneberger, Fischer, and Brox 2015; Wagner et al. 2019; Zhao et al. 2025). In forestry applications, U-Net-based and related architectures have recently been adapted to segment individual tree crowns and discriminate between multiple species from UAV imagery, often achieving high accuracies but typically relying on extensive annotated datasets and carefully designed training pipelines (Huang et al. 2024).

This project aims to develop and evaluate a U-Net-based deep learning model for multi-class semantic segmentation of tree species using a single high-resolution UAV orthomosaic as part of a course on Applied Deep Learning. Specifically, the task involves assigning each pixel to one of several predefined species classes and analysing how segmentation performance is achieved under strong data constraints, such as limited manual annotations and class imbalance.

2. Materials and Methods

Figure 1 provides an overview of the complete pipeline, from UAV data and manual annotation in QGIS, through tiling and mask rasterisation, to U Net training and evaluation.

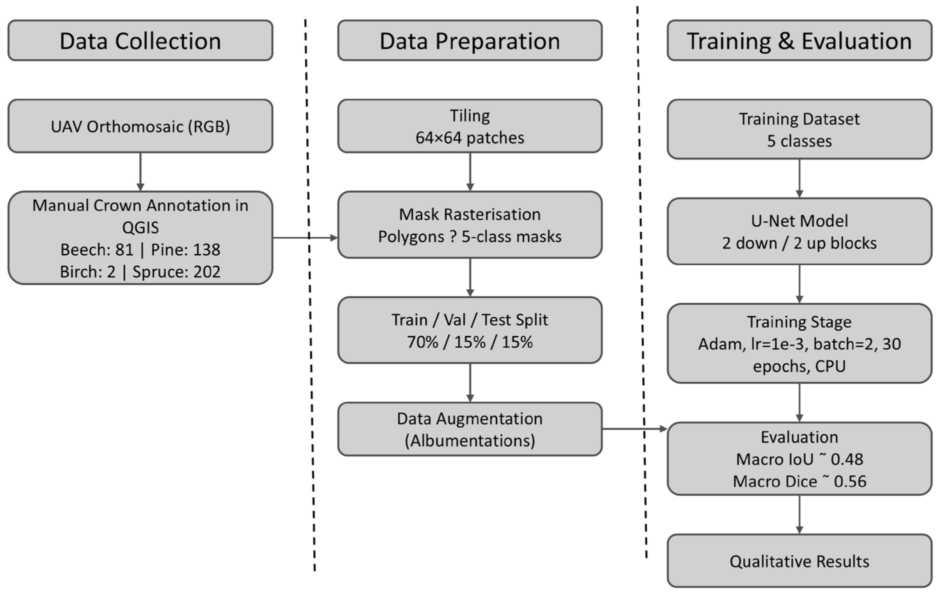

Figure 1: Overview of the data collection, data preparation, and training & evaluation steps used for multi-class tree species segmentation from UAV imagery.

2.1 Data and annotation

The input data consist of a high-resolution RGB orthomosaic of a forested area in Germany downloaded from Zenodo (https://doi.org/10.5281/zenodo.15168163) (Sinan 2022). Tree crowns were manually annotated in QGIS, ensuring polygons did not overlap or leave gaps. Annotated crowns per species: beech (81), pine (138), birch (2), spruce (202), showing strong class imbalance.A Python pipeline (Google Colab) was used to tile the orthomosaic into 256×256 pixel patches, rasterise masks with class IDs, and split tiles into 70/15/15 train/validation/test sets. Coordinate reprojection and border clamping ensured alignment and size consistency.

2.2 Model architecture

A small U-Net-like CNN was used, with two downsampling and two upsampling stages and skip connections. Each block contains two 3×3 convolutions with ReLU activations. The output layer is a 1×1 convolution producing logits for five classes. Cross-entropy loss was used, and predictions are obtained via argmax.

2.3 Training setup

Tiles were normalised; masks used nearest-neighbour resizing to preserve labels. The model was trained with Adam (learning rate 1e-3, batch size 2) for 20 or 30 epochs on CPU (~316 min for 30 epochs). Mean IoU and Dice coefficients per class were computed from confusion matrices on the held-out test set.

2.4 Experiments

The project follows an exploratory "hacking" strategy: start from a simple baseline and iteratively modify the setup. The initial configuration included six classes (background plus five tree species, including locust). A baseline model without augmentation was trained for 20 and then 30 epochs. Next, Albumentations was used to perform random flips and geometric transforms on the tiles, and training was repeated.Because several classes remained poorly predicted, the label space was simplified to five classes by removing locust. The dataset was regenerated using the same tiling pipeline but with updated class IDs, and models were retrained for 20 and 30 epochs. The final configuration used five classes and 30 training epochs.

3. Results

3.1 Quantitative performance

As summarised in Table 1, the six-class configuration reached a macro mean IoU of ~0.36 regardless of augmentation. Simplifying to five classes and training for 30 epochs improved performance to a macro mean IoU of ~0.48 and macro mean Dice of ~0.56. Background, pine, and spruce achieved strong metrics, while beech was moderate and birch nearly zero due to class imbalance.

Table 1. Class-wise Mean IoU and Mean Dice coefficients for the final five-class model (30 epochs)

Class	Mean IoU	Mean Dice
Background	0.9197	0.9580
Beech	0.1472	0.2443
Pine	0.7517	0.8569
Birch	0.0019	0.0038
Spruce	0.5711	0.7125
Macro mean	**≈ 0.48**	**≈ 0.56**

3.2 Training dynamics and qualitative assessment

Training and validation loss curves (Figure 2) decrease steadily over 30 epochs and remain close, indicating stable optimisation and minimal overfitting. Loss reduction slows after ~20 epochs, with 30-epoch models yielding slightly better metrics. Qualitative inspection shows the model delineates major crowns and separates neighbouring trees for dominant species, while dense regions, crown boundaries, and rare species remain challenging. Representative results are available on GitHub, and an interactive interface on Hugging Face allows external users to assess model behaviour.

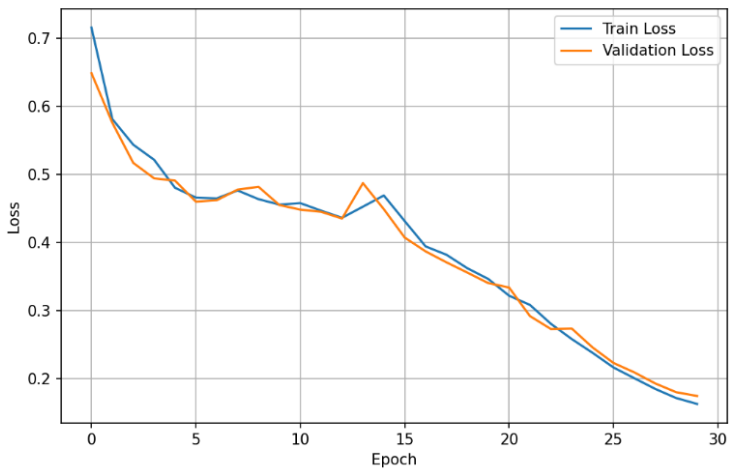

Figure 2: Training and validation loss with 30 epochs.

4. Discussion

The final U-Net demonstrates that even a small model trained on a limited manually annotated dataset can achieve reasonable segmentation for dominant tree species (macro mean IoU ≈0.48). Class-wise variation highlights limitations: birch is poorly predicted and beech moderately, driven by severe class imbalance and visual similarity. Simple geometric augmentation did not compensate for these data limitations, suggesting that class-weighted losses, synthetic oversampling, or species-specific augmentation would likely improve results.

From a deep learning perspective, U-Net is a suitable baseline: simple, CPU-executable, and able to learn coherent crowns from a few hundred tiles. Nevertheless, dataset quality and annotation effort remain the dominant bottleneck. For future iterations, early design of balanced annotation strategies, automated QGIS workflows, and systematic hyperparameter exploration would save time and improve results.

5. Conclusion

This project presented a full pipeline for multi-class segmentation of tree species using UAV orthomosaic imagery and U-Net. Simplifying the label space and refining the training setup allowed the final five-class model to reach a macro mean IoU of ~0.48 and macro Dice of ~0.56, substantially improving over the initial six-class setup. The work highlights that deep learning can effectively segment dominant species with limited data, but balanced annotations and careful dataset design are critical. Future extensions include collecting more diverse data, addressing class imbalance, and moving from tile-based inference to full-scene predictions at forest-stand scale.

Lessons Learned: Annotation is time-consuming, tiling/mask generation requires debugging, Albumentations alone cannot solve class imbalance, CPU training is slow (~316 minutes for 30 epochs), and early planning of dataset strategy would have saved significant time.

6. References

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